Kazim Abbas

Backend/AI engineer

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PROFESSIONAL SUMMARY

Best AI Solution, AI Preneur '26 winner. Backend engineer integrating AI into real systems to automate end-to-end workflows and remove repetitive manual work. Strong understanding of project architecture across voice-first AI, real-time systems, and autonomous pipelines.

HIGHLIGHTS & ACHIEVEMENTS

- Multiple projects shipped across Python, Java, JavaScript, and FullStack stacks; 3 highlighted below.
- **Best AI Solution, AI Preneur '26** (top performer across 200+ participants, multiple institutes).
- **2nd Place, NASA Space Apps Challenge 2025, Karachi** (1,000+ participants, 200+ teams).
- **Open-source publisher:** shipped `fxdc` to PyPI (MIT, 7+ releases Dec 2024 to Aug 2025, current v5.0.0) with a Java parser and VSCode syntax extension for cross-runtime portability.
- **AI automation builder:** autonomous end-to-end video-editing pipeline (Remotion + Python + Fast Whisper) producing 40+ shipped short-form videos, with 50-60/day throughput and 8-15x per-video speed-up over manual editing.

PROJECTS

TeachBack

01/2025 - Present

Best AI Solution, AI Preneur '26

- Built voice-first AI learning SaaS on the **Protege Effect**, winning **Best AI Solution, AI Preneur '26** (200+ participants).
- Architected Django Channels + Daphne ASGI WebSocket pipeline at `/ws/stream/<uuid>/`, streaming 10s chunks with sub-second STT for real-time voice interruption.
- Engineered 3-tier session-state model (Redis 30-min TTL -> PostgreSQL -> in-memory) for sub-1s reconnects and resumable cross-tab conversations.
- Shipped 7-service Docker Compose stack (Postgres, Redis, Django, Celery, React, Nginx) with Stripe billing, JWT auth, and WeasyPrint PDF evaluations.

Tech: Django, Django REST Framework, Django Channels, WebSockets, ASGI/Daphne, PostgreSQL, Redis, Celery, Stripe, JWT, Docker Compose, React, Vite, Tailwind CSS, AI/LLM

FedxD Data Container (FxDC)

12/2024 - 08/2025

- Designed `.fxdc` serialization format (Lexer -> Parser -> FxDCObject + ParseObject round-trip), handling 5+ nesting levels with type metadata preservation.
- Built extensible class system via `Config.add_class` + decorators with FxDCField metadata (type-checking, defaults, constraints).
- Bundled handlers for ~10 native Python types + Pandas DataFrame + NumPy NDArray/Matrix + Datetime, auto-loaded with graceful skip on missing deps.
- Shipped `fxdc` to PyPI (MIT) over 7+ releases (Dec 2024 - Aug 2025, v5.0.0) with Java parser and VSCode syntax extension for cross-runtime portability.

Tech: Python, PyPI Packaging, Java parser, VSCode extension, pytest

AI Video Editing Pipeline

04/2026 - Present

Agentic end-to-end short-form video production, 50-60 finished videos per day

- Built agentic end-to-end video-editing pipeline with 8 autonomous production steps (merge, subtitles, keyword detection, metadata, images, music, Remotion edit, render) and zero human-in-the-loop per video.
- Architected Python orchestrator + Remotion (React) renderer (1080x1920) with 10 composable skills, Faster Whisper transcription, and auto-ducked music at 15-25% during narration.
- Sustained throughput of 50-60 finished videos/day on a 20-24 hr unattended batch, each carrying 8-10 images, Faster Whisper transcript, and SRT subtitles.

- Shipped 40+ medical-reel videos, achieving 8-15x per-video speed-up vs manual editing (20-30 min automated vs 1-6 hr manual) and 10x-30x the daily volume of a single human editor.

Tech: Remotion, React, Python (uv), Fast Whisper (CTranslate2), yt-dlp, ffmpeg, Browser automation, Pipeline orchestration, Short-form video, AI integration

SKILLS

Languages	Backend	Data	Infrastructure
Python, JavaScript, SQL, Bash	Django, DRF, Channels, Celery, WebSockets, REST, JWT, Stripe	PostgreSQL, Redis, SQLite	Docker, Nginx, Linux, Git

AI/Media	Frontend
LLM APIs, Deepgram, Faster Whisper, Remotion, FFmpeg	React, Vite, Tailwind CSS

EXPERIENCE

KayzBlog 07/2024 - Present

Video Editor & Web Developer Karachi, Pakistan

- Built an autonomous AI medical video-editing pipeline (Remotion + Python + Faster Whisper + 10 agentic skills) running 8 production steps end-to-end with zero manual editing, producing 40+ medical-reel videos.
- Automated end-to-end publishing to YouTube (auto metadata, scheduled uploads) and the daily content stack (image research, music, transcription, subtitles), replacing 4-5 hours of manual work per video.
- Built and operate the medical-blog platform on Django (custom admin, 12-15 research articles/month, 5+ updates/month) with the same AI-integration approach applied across the editorial workflow.

Disutils 11/2024 - 01/2025

Lead Developer Remote

- Architected a Discord gaming bot deployed across **10+ servers**, executing **hundreds of commands daily** with minimal downtime.
- Designed database-backed user management, async task handling, and virtual currency systems powering 3+ feature modules (card games, gambling, banking).
- Led 100% of feature design and shipping as Lead Developer, owning the bot's full-stack architecture across API, database, and Discord integration layers.

EDUCATION

Bahria University Karachi Campus 2026 - 2030

Bachelor of Software Engineering (BSE) Karachi, Pakistan

4-year program; expected graduation June 2030.

Fatimiyah Boys College 2024 - 2026

Intermediate, Computer Science Karachi, Pakistan

Happy Home High School (HHS) 2016 - 2024

Matriculation, Computer Science Karachi, Pakistan